Soluções - FME, Volume 06

Capítulo I - Exercícios

1. **a)** $(6+7i)(1+i) = (6,7)(1,1) = (6-7, 6+7) = -1+13i$

b) $(5+4i)(1-i)+(2+i)i = (5,4)(1,-1)+(2,1)(0,1) = (5+4+0-1, -5+4+2-0) = 8+i$

c) $(1+2i)^2-(3+4i) = (1,2)^2-(3,4) = (1-2^2,4)-(3,4) = -6$

2. **a)** $(3+2i)+(2-5i) = 5-3i$

b) $(5-2i)-(2+8i) = 3-10i$

c) $(1+i)+(1-i)-2i = 2-2i$

d) $(6+7i)-(4+2i)+(1-10i) = 3-5i$

3. **a)** $(2-3i)(1+5i) = (2+15, 10-3) = 17+7i$

b) $(1+2i)(2+i) = (2-2, 1+4) = 5i$

c) $(4-3i)(5-i)(1+i) = (20-3, -4-15)(1,1) = (17+19, 17-19) = 36-2i$

d) $(7+2i)(7-2i) = (49+4, -14+14) = 53$

4. $u = 4+3i, v = 5-2i, uv = ?$

$uv = (4,3)(5,-2) = (20+6, -8+15) = 26+7i$

5. **a)** $(3+2i)^2 = (3,2)^2 = (3^2-2^2, 2\cdot 6) = 5+12i$

b) $(5-i)^2 = (5,-1)^2 = (5^2-(-1)^2, -10) = 24-10i$

c) $(1+i)^3 = (1,1)^3 = (1^2-1^2, 2)(1,1) = (0-2, 0+2) = -2+2i$

6. $f(z) = z^2-z+1, f(1-i) = ?$

$f(1-i) = (1,-1)^2-(1,-1)+(1,0) = (0,-2)+(0,1) = -i$

7. $f(z) = z^4+iz^3-(1+2i)z^2+3z+1+3i, f(1+i) = ?$

$f(1+i) = (1,1)^2 = (0,2)$

$= (0,2)^2+(0,1)(0,2)(1,1)-(1,2)(0,2)+(3,3)+(1,3)$

$= (-4,0)+(-2,0)(1,1)-(-4,2)+(4,6)$

$= 2[(-2,0)+(-1,-1)-(-2,1)+(2,3)]$

$= 2(1,1) = 2(1+i)$

8. Prove que $(1+i)^2 = 2i$ e coloque na forma algébrica $z = \dfrac{(1+i)^{80}-(1+i)^{82}}{i^{96}}$

1. $(1,1)^2 = (1-1, 1+1) = 2i$
2. $z = \dfrac{(2i)^{40} - (2i)^{41}}{(i^2)^{48}} = \dfrac{2^{40} - 2^{41} \cdot i}{1} = 2^{40} - 2^{41}i$

9. **a)** $i^{76} = (i^2)^{38} = 1$

b) $i^{110} = (i^2)^{55} = -1$

c) $i^{97} = (i^2)^{48} \cdot i = i$

d) $i^{503} = (i^2)^{251} \cdot i = -i$

10. Prove que $(1-i)^2 = -2i$ e calcule $(1-i)^{96} + (1-i)^{97}$

$(1,-1)^2 = (1-(-1)^2, -1-1) = -2i$

$(1,-1)^{96} + (1,-1)^{97} = (-2i)^{48} + (-2i)^{48}(1,-1) = 2^{48} + 2^{48} - 2^{48}i = 2^{49} - 2^{48}i$

11. Se $i^2 = -1$, calcule $(1+i)^{12} - (1-i)^{12}$

$= (2i)^6 - (-2i)^6 = -2^6 + 2^6 = 0$

12. Simplifique $\dfrac{(2+i)^{101}(2-i)^{50}}{(-2-i)^{100}(i-2)^{49}}$

$= \dfrac{(2,1)^{101}(2,-1)^{50}}{(-(2,1))^{100}(-(2,-1))^{49}} = \dfrac{(2,1)^{101}(2,-1)^{50}}{(2,1)^{100}(-1)(2,-1)^{49}} = -(2,1)(2,-1) = -(4+1,-2+2) = -5$

13. A igualdade $(1+i)^n = (1-i)^n$ é válida para os números naturais divisíveis por qual natural?

- Para $n = 2k, k \in \mathbb{N} \Rightarrow ((1+i)^2)^k = ((1-i)^2)^k \Rightarrow (2i)^k = (-2i)^k \Rightarrow 2^k i^k = (-1)^k 2^k i^k$

  $\Rightarrow (-1)^k = 1 \Rightarrow k = 2k'$

  $\Rightarrow n = 4k'', k'' \in \mathbb{N}$
- Para $n = 2k+1, k \in \mathbb{N} \Rightarrow ((1+i)^2)^k(1+i) = ((1-i)^2)^k(1-i) \Rightarrow (2i)^k(1+i) = (-2i)^k(1-i)$

  $\Rightarrow \dfrac{(1+i)(1+i)}{(1-i)(1+i)} = (-1)^k \Rightarrow (-1)^k = i \implies \perp$

$\therefore$ n deve ser divisível por 4.

14. **a)** $2 + 3yi = x + 9i \quad x = 2, 3y = 9 \Rightarrow \begin{cases} x = 2 \\ y = 3 \end{cases}$

b) $(x+yi)(3+4i) = 7 + 26i \quad (3x-4y, 4x+3y) = (7, 26) \Rightarrow \begin{cases} 3x - 4y = 7 \\ 4x + 3y = 26 \end{cases}$

$\cancel{3x} - 4y - \dfrac{3}{\cancel{4}}\cancel{4x} - \dfrac{3}{4} \cdot 3y = 7 - \dfrac{3}{4} \cdot 26 \Rightarrow -4y - \dfrac{9}{4}y = 7 - \dfrac{3 \cdot 26}{4} \Rightarrow -y\left(\dfrac{4^2 + 3^2}{4}\right) = \dfrac{28 - 78}{4}$

$$y = \frac{50}{25} = 2$$

$$\Rightarrow x = \frac{7 + 4 \cdot 2}{3} = \frac{15}{3} = 5 \Rightarrow \begin{cases} x = 5 \\ y = 2 \end{cases}$$

c) $(x+yi)^2 = 4i \quad (x^2-y^2, 2xy) = (0, 4) \Rightarrow \begin{cases} x^2 = y^2 \Rightarrow x = y \\ xy = 2 \end{cases} \quad \Rightarrow x^2 = 2 \Rightarrow \begin{cases} x = y = \sqrt{2} \\ \text{ou} \\ x = y = -\sqrt{2} \end{cases}$

15. a) $3 + 5ix = y - 15i \Rightarrow \begin{cases} y = 3 \\ 5x = -15 \end{cases} \quad \Rightarrow \begin{cases} x = -3 \\ y = 3 \end{cases}$

b) $(x + yi)(2 + 3i) = 1 + 8i \quad (2x - 3y, 3x + 2y) = (1, 8) \Rightarrow \begin{cases} 2x - 3y = 1 \\ 3x + 2y = 8 \end{cases}$

$$\cancel{2x} - 3y - \frac{2}{\cancel{3}}\cancel{3x} - \frac{2}{3} \cdot 2y = 1 - \frac{2}{3} \cdot 8 \Rightarrow y\left(\frac{-9-4}{3}\right) = \frac{3-16}{3} \Rightarrow y = \frac{-13}{-13} = 1$$

$$\Rightarrow x = \tfrac{1+3}{2} \Rightarrow \begin{cases} x = 2 \\ y = 1 \end{cases}$$

c) $(3 + yi) + (x - 2i) = 7 - 5i \Rightarrow \begin{cases} 3 + x = 7 \\ y - 2 = -5 \end{cases} \quad \Rightarrow \begin{cases} x = 4 \\ y = -3 \end{cases}$

d) $(x + yi)^2 = 2i \Rightarrow (x^2 - y^2, 2xy) = (0, 2) \Rightarrow \begin{cases} x^2 = y^2 \\ xy = 1 \end{cases} \quad \Rightarrow \begin{cases} x = 1 \text{ e } y = 1 \\ \text{ou} \\ x = -1 \text{ e } y = -1 \end{cases}$

e) $(2 - x + 3y) + 2yi = 0 \Rightarrow \begin{cases} 2 - x + 3y = 0 \\ 2y = 0 \end{cases} \quad \Rightarrow \begin{cases} x = 2 \\ y = 0 \end{cases}$

f) $(3 - i)(x + yi) = 20 \quad (3x + y, 3y - x) = (20, 0) \Rightarrow \begin{cases} 3x + y = 20 \\ 3y = x \end{cases}$

$$\Rightarrow 3x + \frac{x}{3} = 20 \Rightarrow 9x + x = 3 \cdot 20 \Rightarrow x = 6 \Rightarrow \begin{cases} x = 6 \\ y = 2 \end{cases}$$

16. $\begin{cases} x + yi = i \\ xi + y = 2i - 1 \end{cases}$

$$x = i(1 - y) \Rightarrow (1 - y)i^2 + y = 2i - 1 \Rightarrow y - 1 + y = 2i - 1 \Rightarrow y = i$$

$$\Rightarrow x + i^2 = i \Rightarrow x = 1 + i \Rightarrow \begin{cases} x = 1 + i \\ y = i \end{cases}$$

17. $(a, b)(c, d) = (ac - bd, ad + bc) = (x, 0), x \in \mathbb{R}$

$\Rightarrow ad + bc = 0 \Rightarrow ad = -bc \Rightarrow \frac{a}{c} = -\frac{b}{d}$

18. $(a, b)^2 \Rightarrow (a^2 - b^2, 2ab)^2 = ((a^2 - b^2)^2 - 4a^2b^2, 4ab(a^2 - b^2))$

$\Rightarrow$ 1. $4ab(a^2 - b^2) = 0 \Rightarrow a = \pm b \ (a \neq 0 \wedge b \neq 0)$ ou $a = 0$ ou $b = 0$

$\Rightarrow$ 2. $(a^2 - b^2)^2 - 4a^2b^2 < 0 \Rightarrow (a^2 - b^2)^2 < 4a^2b^2$

- $a = 0, b \neq 0 \Rightarrow (-b^2)^2 < 0 \Rightarrow b^4 < 0 \Rightarrow \perp$
- $a \neq 0, b = 0 \Rightarrow a^4 < 0 \Rightarrow \perp$
- $a = \pm b, a \neq 0, b \neq 0 \Rightarrow ((\pm b)^2 - b^2)^2 < 4(\pm b)^2b^2 \Rightarrow 0 < 4b^4, \forall b \neq 0$

$\therefore \begin{cases} a = \pm b \\ ab \neq 0 \end{cases}$

19. **a)** $\frac{2}{i} \cdot \frac{-i}{-i} = -2i$

b) $\frac{3}{2+i} \cdot \frac{2-i}{2-i} = \frac{6}{5} - \frac{3}{5}i$

c) $\frac{1+2i}{3-i} \cdot \frac{3+i}{3+i} = \frac{1}{10} + \frac{7}{10}i$

d) $\frac{i^9}{4-3i} \cdot \frac{4+3i}{4+3i} = \frac{i^8 \cdot i(4+3i)}{25} = \frac{-3}{25} + \frac{4}{25}i$

20. **a)** $\frac{1}{i} \cdot \frac{i}{i} = -i$

b) $\frac{1}{1+i} \cdot \frac{1-i}{1-i} = \frac{1}{2} - \frac{1}{2}i$

c) $\frac{3+4i}{2-i} \cdot \frac{2+i}{2+i} = \frac{2}{3} + \frac{11}{5}i$

d) $\frac{1+i}{1-i} \cdot \frac{1+i}{1+i} = \frac{2i}{2} = i$

e) $\frac{i^{11} + 2i^{13}}{i^{18} - i^{37}} = \frac{i^{11}(1 + 2i^2)}{i^{18}(1 - i^19)} = \frac{-1}{i^4i^2i(1 - i^{16}i^2i)} = \frac{1}{i(1+i)} = \frac{1}{-1+i} \cdot \frac{-1+i}{-1+i} = -\frac{1}{2} - \frac{1}{2}i$

f) $\frac{1-3i}{3-i} \cdot \frac{3+i}{3+i} = \frac{3}{5} - \frac{4}{5}i$

g) $\frac{i^3 - i^2 + i^{17} - i^{35}}{i^{16} - i^{13} + i^{30}} = \frac{i^2(i - 1 + i^{12}i^3 - i^{32}i)}{i^{13}(i^3 - 1 + i^{17})} = \frac{i - 1 - i - i}{i^3(-i - 1 + i^{16}i)} = \frac{-(1+i)}{-i(-1)} \cdot \frac{-i}{-i} = -1 + i$

h) $\frac{1+i}{(1-i)^2} = \frac{1+i}{-2i} \cdot \frac{2i}{2i} = \frac{-2+2i}{4} = -\frac{1}{2} + \frac{1}{2}i$

21. $\frac{1+3i}{2-i} \cdot \frac{2+i}{2+i} = -\frac{1}{5} - \frac{7}{5}i$

22. $\frac{1+i}{i} \cdot \frac{-i}{-i} = 1 - i \implies 1 + i$

23. $z = \frac{1-i}{1+i} \cdot \frac{-2i}{2} = -i \implies z^{-1} = \frac{1}{-i} \cdot \frac{i}{i} = i \implies \overline{z^{-1}} = -i$

24. $u^2 - v^2 = 6, \overline{u} + \overline{v} = 1 - i, u - v = ?$

$(u - v)(u + v) = 6 \implies u - v = \frac{6}{u+v}$

$\overline{u} + \overline{v} = 1 - i \implies \overline{\overline{u} + \overline{v}} = \overline{1 - i} \implies u + v = 1 + i$

$\therefore u - v = \frac{6}{u+v} = \frac{6}{1+i} \cdot \frac{1-i}{1-i} = 3 - 3i$

25. Provavelmente tem uma informação errada na questão $u = 1$

26. $\left(\frac{1+i}{1-i}\right)^n, n \in \mathbb{Z}$

$\left(\frac{1+i}{1-i} \cdot \frac{1+i}{1+i}\right)^n = \left(\frac{2i}{2}\right)^n = i^n \therefore 1, i, -1, -i$

27. $\frac{a+bi}{c+di}, c + di \neq 0$

$\frac{a+bi}{c+di} \cdot \frac{c-di}{c-di} = \frac{ac - bd + i(ad + bc)}{c^2 + d^2}$

$\implies ax - bd = 0$

$\implies ad + bc = 0$

28. $u = x + yi, v = \frac{1}{2} - \frac{\sqrt{3}}{2}i, \mathrm{Re}(v\overline{u}) = ?$

$v\overline{u} = (x + yi)(\frac{1}{2} + \frac{\sqrt{3}}{2}) = (\frac{x}{2} - \frac{\sqrt{3}}{2}y, \frac{\sqrt{3}}{2}x + \frac{y}{2})$

$\mathrm{Re}(v\overline{u}) = \frac{x}{2} - \frac{\sqrt{3}}{2}y$

29. $z = x + iy, x^2 + y^2 \neq 0, \mathrm{Im}(z + \frac{1}{z}) = 0, z = ?$

$z + \frac{1}{z} = \frac{(x+iy)(x+iy)+1}{x+iy} \cdot \frac{(x-iy)}{(x-iy)} = \frac{(x-iy)(x^2 - y^2 + 2xyi + 1)}{x^2 + y^2}$

$= \frac{(x,-y)}{x^2+y^2}(x^2 - y^2 + 1, 2xy) = \frac{(x(x^2 - y^2 + 1) + 2xy^2, 2x^2y - y(x^2 - y^2 + 1),)}{x^2 + y^2}$

$\implies Im(z + \frac{1}{z}) = 2x^2y - y(x^2 - y^2 + 1) = 0 \implies 2x^2y = y(x^2 - x^2 + 1)$

$\implies 2x^2 = x^2 - y^2 + 1 \implies x^2 + y^2 = 1$

30. $z_1, z_2 \in \mathbb{C}, \text{Im}(z_1 + z_2) = \text{Im}(z_1 z_2) = 0$

$$z_1 = a + bi, z_2 = c + di \implies \begin{cases} z_1 z_2 = (ac - bd, ad + bc) \\ z_1 + z_2 = (a + c, b + d) \end{cases} \implies \begin{cases} ad + bc = 0 \\ b + d = 0 \end{cases}$$

$$\implies \begin{cases} ad = -bc \\ b = -d \end{cases} \implies \begin{cases} a = c \\ b = -d \end{cases} \implies z_1 = \overline{z}_2 \text{ ou } z_1, z_2 \in \mathbb{R}$$

31. $z = \dfrac{2 - xi}{1 + 2xi}, x \in \mathbb{R}, \text{Re}(z) = 0$

$$z = \frac{2 - xi}{1 + 2xi} \cdot \frac{1 - 2xi}{1 - 2xi} = \frac{(2x - 2x^2) + (-4x - x)i}{1 + 4x^2}$$

$$z = \frac{2(1 - x^2) - 3xi}{4x^2 + 1} \implies \frac{2(1 - x^2)}{4x^2 + 1} = 0 \implies 1 - x^2 = 0 \implies x = \pm 1$$

32. $z = \dfrac{1 + 2i}{2 + ai}, a = ?$

$$\frac{1 + 2i}{2 + ai} \cdot \frac{2 - ai}{2 - ai} = \frac{(2 + 2a) + (-a + 4)i}{a^2 + 4} \implies \frac{-a + 4}{a^2 + 4} = 0 \implies a = 4$$

33. $\text{Im}(z(5 + 8i)) = 0, \text{Re}(\dfrac{z}{1 + i}) = 0, z = ?$

$$(x + yi)(5 + 8i) = (5x - 8y) + (8x + 5y)i \implies 8x + 5y = 0$$

$$\frac{x + yi}{1 + i} \cdot \frac{1 - i}{1 - i} = \frac{(x + y) + (-x + y)i}{2} \implies \frac{x + y}{2} = 0 \implies x + y = 0$$

$$\implies \perp$$

34. $\dfrac{z}{1 - i} + \dfrac{z - 1}{1 + i} = \dfrac{5}{2} + \dfrac{5}{2}i, z = ?$

$$\frac{z(1 + i) + (z - 1)(1 - i)}{(1 - i)(1 + i)} = \frac{5}{2}(1 + i) \implies z + \cancel{zi} + z - \cancel{zi} - 1 + i = \frac{5}{\cancel{2}} \cdot \cancel{2}(1 + i)$$

$$2z - 1 + i = 5(1 + i) \implies 2z = 5 + 5i + 1 - i = 6 + 4i \implies z = 3 + 2i$$

35. $z \in \mathbb{C}, \overline{z} = -2zi$

$$z = x + yi \implies x - yi = -2i(x + yi) = -2xi + 2y \implies \begin{cases} x = 2y \\ y = 2x \end{cases} \implies z = 0$$

36. $z = x + yi, u = \dfrac{1}{2} - \dfrac{\sqrt{3}}{2}i, \text{Re}(u\overline{z}) = ?, \text{Im}(u\overline{z}) = ?$

$$u\overline{z} = \left(\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)(x - yi) = \left(\frac{x}{2} - \frac{\sqrt{3}}{2}y, \frac{-y}{2} - \frac{\sqrt{3}}{2}x\right)$$

$$\begin{cases} \mathrm{Re}(u\bar{z}) = \dfrac{x}{2} - \dfrac{\sqrt{3}}{2}y \\ \mathrm{Im}(u\bar{z}) = \dfrac{-\sqrt{3}}{2}x - \dfrac{y}{2} \end{cases}$$

37. Prove que $\overline{z^n} = (\bar{z})^n, \forall n \in \mathbb{N}$.

$$z = a + bi \implies z^n = (a+bi)^n = \sum_k^n \binom{n}{k} a^k (bi)^{n-k}$$

$$\implies \overline{z^n} = \sum_k^n \binom{n}{k} a^k (-bi)^{n-k} = (a + (-bi))^n = (\bar{z})^n$$

38. Prove que se a equação $x^2 + (a+bi)x + (c+di) = 0$, com $a, b, c, d \in \mathbb{R}$, admite uma raiz real então $abd = d^2 + b^2 c$.

$$\exists x \in \implies x^2 + ax + bxi + c + di = 0 \implies (x^2 + ax + c, xb + d) = 0 \implies \begin{cases} x^2 + ax + c = 0 \\ xb + d = 0 \end{cases}$$

$$\implies x = \frac{-d}{b} \implies \frac{d^2}{b^2} + \frac{a(-d)}{b} + c = 0 \implies d^2 - abd + cb^2 = 0 \implies abd = d^2 + cb^2$$

39. $z\bar{z} + (z - \bar{z}) = 13 + 6, z = ?$

$$x^2 + y^2 + \cancel{x} + yi - \cancel{x} + yi = 13 + 6i \implies x^2 + y^2 + 2yi = 13 + 6i \implies \begin{cases} x^2 + y^2 = 13 \\ 2y = 6 \end{cases}$$

$$\implies x^2 = 13 - 9 = 4 \implies \begin{cases} x = \pm 2 \\ y = 3 \end{cases}$$

40. $z \in \mathbb{C} | z^3 = \bar{z}$

$$(x^2 - y^2, 2xy)(x, y) = (x^2 - xy^2 - 2xy^2, x^2 y - y^3 + 2x^2 y) = (x, -y)$$

$$\implies (x^3 - 3xy^2, y^3 - 3x^2 y) = (x, y) \implies \begin{cases} x(x^2 - 3y^2 - 1) = 0 \\ y(y^2 - 3x^2 - 1) = 0 \end{cases}$$

$$\implies 1.\ x = y = 0 \implies z = 0$$

$$\implies 2.\ x \neq 0 \wedge y = 0 \implies x^2 - 3y^2 - 1 = 0 \implies x = \pm 1 \implies z = \pm 1$$

$$\implies 3.\ x = 0 \wedge y \neq 0 \implies x^2 - 3y^2 - 1 = 0 \implies y = \pm 1 \implies z = \pm i$$

$$\implies 4.\ x \neq 0 \wedge y \neq 0 \implies \begin{cases} x^2 - 3y^2 - 1 = 0 \\ y^2 - 3x^2 - 1 = 0 \end{cases} \implies y^2 - 3(1 + 3y^2) - 1 = 0$$

$$\implies -8y^2 = 4 \implies y \in \mathbb{C} \implies \perp$$

41. $z \in \mathbb{R} | z^2 = i$

$$(x^2 - y^2, 2xy) = (0, 1) \implies \begin{cases} x^2 = y^2 \\ 2xy = 1 \end{cases} \implies \begin{cases} x = \pm y \\ xy = \frac{1}{2} \end{cases}$$

$$\implies \begin{cases} x = y \implies x^2 = \dfrac{1}{2} \implies x = y = \pm\dfrac{\sqrt{2}}{2} \\ x = -y \implies x^2 = -\dfrac{1}{2} \implies x \in \mathbb{C} \implies \perp \end{cases}$$

42. $z \in \mathbb{C} | z^2 = 1 + i\sqrt{3}$

$$(x^2 - y^2, 2xy) = (1, \sqrt{3}) \implies \begin{cases} x^2 = 1 + y^2 \\ 2xy = \sqrt{3} \implies 4x^2y^2 = 3 \end{cases} \implies 4x^2y^2 = 4y^2(1 + y^2)$$

$$\implies y^4 + y^2 - \frac{3}{4} = 0 \implies k^2 + k - \frac{3}{4} = 0 \quad (k = y^2) \implies k = \frac{-1 \pm \sqrt{1 - \cancel{4} \cdot \frac{3}{\cancel{4}}}}{2} = \begin{cases} \dfrac{1}{2} \\ \dfrac{-3}{2} \end{cases}$$

$$\implies y = \begin{cases} \pm\dfrac{\sqrt{2}}{2} \implies \begin{cases} y = \dfrac{\sqrt{2}}{2} \implies x = \dfrac{\sqrt{6}}{2} \\ y = -\dfrac{\sqrt{2}}{2} \implies x = -\dfrac{\sqrt{6}}{2} \end{cases} \\ \pm\sqrt{\dfrac{-3}{2}} \implies y \in \mathbb{C} \implies \perp \end{cases}$$

$$\therefore z = \frac{\sqrt{6}}{2} + \frac{\sqrt{2}}{2}i \text{ ou } z = -\frac{\sqrt{6}}{2} - \frac{\sqrt{2}}{2}i$$

43. Prove que $x^2 + y^2 = 1 \implies \dfrac{1 + x + iy}{1 + x - yi} = x + iy.$

$$\frac{1 + x + iy}{1 + x - iy} \cdot \frac{1 + x + iy}{1 + x + iy} = \frac{(1 + x, y)(1 + x, y)}{(1 + x, -y)(1 + x, y)} = \frac{(x^2 + 2x + 1 - y^2, 2y(x + 1))}{x^2 + 2x + 1 + y^2}$$

$$= \frac{(x^2 + 2x + x^2, 2y(x + 1))}{2x + 2} = \frac{2x(x + 1), 2y(x + 1)}{2(x + 1)} = x + iy$$

44. Prove que $\dfrac{1 + \sin x + i\cos x}{1 - \sin x - i\cos x} = (\tan x + \sec x)i, \forall x \neq \dfrac{\pi}{2} + k\pi.$

$$\frac{\frac{1}{\cos x} + \frac{\sin x}{\cos x} + i\frac{\cos x}{\cos x}}{\frac{1}{\cos x} - \frac{\sin x}{\cos x} - i\frac{\cos x}{\cos x}} = \frac{\sec x + \tan x + i}{\sec x - \tan x - i} \cdot \frac{\sec x - \tan x + i}{\sec x - \tan x + i}$$

$$= \frac{(\sec^2 x - \tan^2 x - 1, \sec x + \tan x + \sec x - \tan x)}{\sec^2 - 2\sec x \tan x + \tan^2 x + 1} = \frac{(0, 2\sec x)}{2\sec x(\sec x - \tan x)} \cdot \frac{\sec x + \tan x}{\sec x + \tan x}$$

$$= (\tan x + \sec x)i$$

45. **a)** 4

$$\begin{cases} \rho = \sqrt{4^2} = 4 \\ \cos\theta = 1 \\ \sin\theta = 0 \end{cases} \implies 4 = 4(\cos 0 + i \sin 0)$$

b) $1 + i\sqrt{3}$

$$\begin{cases} \rho = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{4} = 2 \\ \cos\theta = \dfrac{1}{2} \\ \sin\theta = \dfrac{\sqrt{3}}{2} \end{cases} \implies 1 + i\sqrt{3} = 2(\cos\frac{\pi}{3} + i \sin\frac{\pi}{3})$$

c) $3i$

$$\begin{cases} \rho = \sqrt{3^2} = 3 \\ \cos\theta = 0 \\ \sin\theta = \dfrac{3}{3} = 1 \end{cases} \implies 3i = 3(\cos\frac{\pi}{2} + i \sin\frac{\pi}{2})$$

d) $\sqrt{2} + i\sqrt{2}$

$$\begin{cases} \rho = (\sqrt{2})^2 + (\sqrt{2})^2 = \sqrt{4} = 2 \\ \cos\theta = -\dfrac{\sqrt{2}}{2} \\ \sin\theta = \dfrac{\sqrt{2}}{2} \end{cases} \implies \sqrt{2} + i\sqrt{2} = 2(\cos\frac{3\pi}{4} + i \sin\frac{3\pi}{4})$$

e) -5

$$\begin{cases} \rho = \sqrt{5^2} = 5 \\ \cos\theta = 1 \\ \sin\theta = 0 \end{cases} \implies -5 = 5(\cos\pi + i \sin\pi)$$

f) $-2i$

$$\begin{cases} \rho = \sqrt{2^2} = 2 \\ \cos\theta = 0 \\ \sin\theta = \dfrac{-2}{2} = -1 \end{cases} \implies -2i = 2(\cos\frac{3\pi}{2} + i \sin\frac{3\pi}{2})$$

g) $-5 - 5i$

$$\begin{cases} \rho = \sqrt{5^2 + 5^2} = \sqrt{2 \cdot 5^2} = 5\sqrt{2} \\ \cos\theta = \dfrac{-5}{5\sqrt{2}} = -\dfrac{\sqrt{2}}{2} \\ \sin\theta = \dfrac{-5}{5\sqrt{2}} = -\dfrac{\sqrt{2}}{2} \end{cases} \implies -5 - 5i = 5\sqrt{2}(\cos\frac{5\pi}{4} + i \sin\frac{5\pi}{4})$$

h) $2 - 2i$

$$\begin{cases} \rho = \sqrt{2^2 + 2^2} = \sqrt{2^3} = 2\sqrt{2} \\ \cos\theta = \dfrac{2}{2\sqrt{2}} = \dfrac{\sqrt{2}}{2} \\ \sin\theta = -\dfrac{2}{2\sqrt{2}} = -\dfrac{\sqrt{2}}{2} \end{cases} \implies 2 - 2i = 2\sqrt{2}(\cos\frac{7\pi}{4} + i\sin\frac{7\pi}{4})$$

46. **a)** $|3 - 4i| = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$

b) $|\sqrt{2} - i\sqrt{2}| = \sqrt{(\sqrt{2})^2 + (\sqrt{2})^2} = \sqrt{2 + 2} = \sqrt{4} = 2$

c) $|12 + 5i| = \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = 13$

d) $|\cos\theta + i\sin\theta| = \sqrt{\cos^2\theta + \cos^2\theta} = \sqrt{1} = 1$

e) $|\tan\theta + i| = \sqrt{\tan^2\theta + 1^2} = \sqrt{\sec^2\theta} = |\sec\theta|$

f) $|24 + 7i| = \sqrt{24^2 + 7^2} = \sqrt{625} = 25$

47. $\left|\dfrac{a+bi}{a-bi}\right| = \left|\dfrac{a+bi}{a-bi} \cdot \dfrac{a+bi}{a+bi}\right| = \dfrac{|a^2 - b^2 + 2abi|}{a^2 + b^2} = \dfrac{\sqrt{(a^2-b^2)^2 + (2ab)^2}}{a^2 + b^2}$

$= \dfrac{\sqrt{a^4 - 2a^2b^2 + b^4 + 4a^2b^2}}{a^2 + b^2} = \dfrac{\sqrt{(a^2)^2 + 2a^2 + b^2 + (b^2)^2}}{a^2 + b^2} = \dfrac{\sqrt{(a^2+b^2)^2}}{a^2 + b^2} = \dfrac{a^2 + b^2}{a^2 + b^2} = 1$

48. $\left|\dfrac{1}{1 + i\tan x}\right|, x \neq \dfrac{\pi}{2} + k\pi$

$\left|\dfrac{1}{1 + i\tan x} \cdot \dfrac{1 - i\tan x}{1 - i\tan x}\right| = \left|\dfrac{1 - i\tan x}{1^2 + \tan^2 x}\right| = \dfrac{|1 - i\tan x|}{\sec^2 x}$

$= \dfrac{\sqrt{1^2 + \tan^2 x}}{\sec^2 x} = \dfrac{\sqrt{\sec^2 x}}{\sec^2 x} = \dfrac{1}{\sec x} = \cos x$

49. **a)** $3 + 3i$

$$\begin{cases} \rho = \sqrt{3^2 + 3^2} = \sqrt{2 \cdot 3^2} = 3\sqrt{2} \\ \cos\theta = \dfrac{3}{3\sqrt{2}} = \dfrac{\sqrt{2}}{2} \\ \sin\theta = \dfrac{3}{3\sqrt{2}} = \dfrac{\sqrt{2}}{2} \end{cases} \implies 3 + 3i = 3\sqrt{2}(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4})$$

b) $5 - i5\sqrt{3}$

$$\begin{cases} \rho = \sqrt{5^2 + 5^2\sqrt{3^2}} = \sqrt{25 + 3 \cdot 25} = \sqrt{100} = 10 \\ \cos\theta = \dfrac{5\sqrt{3}}{10} = -\dfrac{\sqrt{3}}{2} \\ \sin\theta = \dfrac{5}{10} = \dfrac{1}{2} \end{cases} \implies 3 - i5\sqrt{3} = 10(\cos\frac{\pi}{4} + i\sin\frac{5\pi}{3})$$

c) $-8 - 8i = -8(1 + i)$

$$\begin{cases} \rho = \sqrt{1^2+1^2} = \sqrt{2} \\ \cos\theta = -\dfrac{\sqrt{2}}{2} \\ \sin\theta = -\dfrac{\sqrt{2}}{2} \end{cases} \implies -8-8i = 8\sqrt{2}(\cos\frac{5\pi}{4} + i\sin\frac{5\pi}{4})$$

d) 11

$$\begin{cases} \rho = \sqrt{11^2} = 11 \\ \cos\theta = \dfrac{11}{11} = 1 \\ \sin\theta = \dfrac{0}{11} = 0 \end{cases} \implies 11 = 11(\cos 0 + i\sin 0)$$

e) $2i$

$$\begin{cases} \rho = \sqrt{2^2} = 2 \\ \cos\theta = \dfrac{0}{2} = 0 \\ \sin\theta = \dfrac{2}{2} = 1 \end{cases} \implies 2i = 2(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2})$$

f) $i^3 = -i$

$$\begin{cases} \rho = \sqrt{1^2} = 1 \\ \cos\theta = \dfrac{0}{1} \\ \sin\theta = \dfrac{-1}{1} = -1 \end{cases} \implies i^3 = 1(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2})$$

g) $-\sqrt{3}+i$

$$\begin{cases} \rho = \sqrt{(\sqrt{3})^2+1^2} = \sqrt{4} = 2 \\ \cos\theta = -\dfrac{\sqrt{3}}{2} \\ \sin\theta = \dfrac{\sqrt{2}}{2} \end{cases} \implies -\sqrt{3}+i = 2(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6})$$

h) $i(1+i) = -1+i$

$$\begin{cases} \rho = \sqrt{1^2+1^2} = \sqrt{2} \\ \cos\theta = -\dfrac{\sqrt{2}}{2} \\ \sin\theta = \dfrac{\sqrt{2}}{2} \end{cases} \implies i(1+i) = \sqrt{2}(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4})$$

i) $2i(1-i) = 2+2i$

$$\begin{cases} \rho = \sqrt{2^2+2^2} = \sqrt{8} = 2\sqrt{2} \\ \cos\theta = \dfrac{\sqrt{2}}{2} \\ \sin\theta = \dfrac{\sqrt{}}{2} \end{cases} \implies 2i(1-i) = 2\sqrt{2}(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4})$$

50. a) $3(\cos\pi + i\sin\pi)$

$$\begin{cases} x = 3\cos\pi = -3 \\ y = 3\sin\pi \end{cases} \implies -3$$

b) $2(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4})$

$$\begin{cases} x = 2\cos\frac{\pi}{4} = 2\cdot\frac{\sqrt{2}}{2} = \sqrt{2} \\ y = 2\sin\frac{\pi}{4} = 2\cdot\frac{\sqrt{2}}{2} = \sqrt{2} \end{cases} \implies \sqrt{2} + i\sqrt{2}$$

c) $4(\cos\frac{11\pi}{6} + i\sin\frac{11\pi}{6})$

$$\begin{cases} x = 4\cos\frac{11\pi}{6} = 4\cos(2\pi - \frac{\pi}{6}) = 4\cdot\frac{\sqrt{3}}{2} = 2\sqrt{3} \\ y = 4\sin\frac{11\pi}{6} = 4\sin(2\pi - \frac{\pi}{6}) = -4\cdot\frac{1}{2} = -2 \end{cases} \implies 2\sqrt{3} - 2i$$

d) $4(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2})$

$$\begin{cases} x = 5\cdot 0 = 0 \\ y = 5\cdot -1 = -5 \end{cases} \implies -5i$$

51. a) $|(1-i)(2-2i)| = |2(1-i)^2| = |2\cdot 2i| = |4i| = 4$

b) $|(1+i\sqrt{3})^6| = (|1+i\sqrt{3}|)^6 = (\sqrt{1^2+(\sqrt{3})^2})^6 = 2^6 = 64$

c) $\left|\frac{3+3i}{1+2i}\right| = \frac{|3+3i|}{|1+2i|} = \frac{\sqrt{3^2+3^2}}{\sqrt{1^2+2^2}} = \frac{3\sqrt{2}}{\sqrt{5}}\cdot\frac{\sqrt{5}}{\sqrt{5}} = \frac{3\sqrt{10}}{5}$

52. a) $|1+i||3| = 3\sqrt{1^2+1^2} = 3\sqrt{2}$

b) $|1-i|^4 = |\sqrt{2}|^4 = 2^2 = 4$

c) $|5+12i||i| = \sqrt{5^2+12^2} = \sqrt{169} = 13$

d) $|1+i||2+2i||4+4i| = \sqrt{2}\cdot\sqrt{8}\cdot\sqrt{32} = \sqrt{2\cdot 2^3\cdot 2^5} = \sqrt{2^8\cdot 2} = 2^4\sqrt{2} = 16\sqrt{2}$

e) $\frac{|1+i|}{|2-2i|} = \frac{\sqrt{2}}{\sqrt{8}}\cdot\frac{\sqrt{8}}{\sqrt{8}} = \frac{\sqrt{2^4}}{8} = \frac{4}{8} = \frac{1}{2}$

f) $\frac{|5i|}{|3+4i|} = \frac{5}{\sqrt{3^2+4^2}} = \frac{5}{\sqrt{25}} = 1$

53. a) $\left(\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)^2 = \frac{1}{2}(1+i\sqrt{3})^2 = \frac{1}{4}(-2+2i\sqrt{3}) = \frac{-1}{2} + \frac{\sqrt{3}}{2}$

$$\begin{cases} \rho = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} = 1 \\ \cos\theta = \frac{-1}{2} \\ \sin\theta = \frac{\sqrt{3}}{2} \end{cases} \implies 1(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3})$$

b) $\frac{1+i}{i} = 1 - i$

$$\begin{cases} \rho = \sqrt{1^2 + 1^2} = \sqrt{2} \\ \cos\theta = \frac{\sqrt{2}}{2} \\ \sin\theta = \frac{-\sqrt{2}}{2} \end{cases} \implies \sqrt{2}(\cos\frac{7\pi}{4} + i\sin\frac{7\pi}{4})$$

c) $5(\cos 30^\circ - i\sin 30^\circ) = 5(\cos\frac{\pi}{6} + i\sin\frac{-\pi}{6})$

$\implies 5(\cos\frac{11\pi}{6} + i\sin\frac{11\pi}{6})$

54. $\frac{1}{1-i} - \frac{1}{i} = \frac{i-1+i}{1+i} \cdot \frac{1-i}{1-i} = \frac{1+3i}{2}$

$\implies \frac{1}{2} + \frac{3}{2}i$

$$\begin{cases} \rho = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{3}{2}\right)^2} = \sqrt{\frac{1}{4} + \frac{9}{4}} = \sqrt{\frac{10}{4}} = \frac{\sqrt{10}}{2} \\ \cos\theta = \frac{1}{2} \cdot \frac{3}{\sqrt{10}} = \frac{\sqrt{10}}{10} \\ \sin\theta = \frac{3}{2} \cdot \frac{2}{\sqrt{10}} = \frac{3\sqrt{10}}{10} \end{cases} \implies \tan\theta = \frac{3\sqrt{10}}{10} \cdot \frac{10}{\sqrt{10}} = 3$$

$\implies \frac{\sqrt{10}}{2}(\cos(\text{atan } 3) + i\sin(\text{atan } 3))$

55. $\frac{1}{1+i\sqrt{3}} \cdot \frac{1-\sqrt{3}}{1-\sqrt{3}} = \frac{1-\sqrt{3}}{1+(\sqrt{3})^2} = \frac{1}{4} - \frac{\sqrt{3}}{4}i$

$$\begin{cases} \rho = \sqrt{\frac{1}{4^2} + \frac{(\sqrt{3})^2}{4^2}} = \sqrt{\frac{1}{16} + \frac{3}{16}} = \sqrt{\frac{1}{4}} = \frac{1}{2} \\ \cos\theta = \frac{1}{2} \\ \sin\theta = \frac{-\sqrt{3}}{2} \end{cases} \implies \frac{1}{2}(\cos\frac{11\pi}{6} + i\sin\frac{11\pi}{6})$$

56. a) $\frac{5+5i}{2-2i} = \frac{5}{2}\left(\frac{1+i}{1-i}\right)\left(\frac{1+i}{1+i}\right) = \frac{10i}{2\cdot 2} = \frac{5}{4}i$

$$\begin{cases} \rho = \frac{5}{4} \\ \cos\theta = 0 \\ \sin\theta = 1 \end{cases} \implies \frac{5}{2}(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2})$$

b) $\frac{1}{i} + (2-i) = \frac{2(1+i)}{i} \cdot \frac{-i}{-i} = 2(1-i)$

$$\begin{cases} \rho = \sqrt{2^2(1^2+1^2)} = 2\sqrt{2} \\ \cos\theta = \frac{\sqrt{2}}{2} \\ \sin\theta = \frac{-\sqrt{2}}{2} \end{cases} \implies 2\sqrt{2}(\cos\frac{7\pi}{4} + i\sin\frac{7\pi}{4})$$

c) $\frac{(1+i\sqrt{3})^3}{i^5} = \frac{(1+i\sqrt{3})^2(1+i\sqrt{3})}{i} \cdot \frac{-i}{-i} = (-2, 2\sqrt{3})(1, \sqrt{3})(-i) = 8i$

$$\begin{cases} \rho = 8 \\ \cos\theta = 0 \\ \sin\theta = 1 \end{cases} \implies 8(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2})$$

57. $z, w \in \mathbb{C}, |z| = |w| = 1, 1 + zw \neq 0, \frac{z+w}{1+zw}$?

$$\frac{z+w}{1+zw} \cdot \frac{1+\overline{zw}}{1+\overline{zw}} = \frac{z + z\overline{zw} + w + z\overline{w}w}{1 + \overline{zw} + zw + z\overline{z}w\overline{w}} = \frac{z + \overline{w} + w + \overline{z}}{1 + \overline{zw} + zw + 1} = \frac{\mathrm{Re}(z) + \mathrm{Re}(w)}{1 + \mathrm{Re}(zw)}$$

$$\therefore \frac{z+w}{1+zw} \in \mathbb{R}$$

58. $z_1 = \rho(\cos\phi + i\sin\phi), z_2 = \rho(\sin\phi + i\cos\phi), z_1 - iz_2 \in \mathbb{R}, \mathbb{C}, \mathbb{I}$?

$$z_1 i - iz_2 = \rho(\cos\phi + i\sin\phi) - i\rho(\sin\phi + i\cos\phi) = \rho(\cos + i\sin\phi - i\sin\phi - i^2\cos\phi)$$

$$= 2\rho\cos\phi \implies z_1 - i_2 \in \mathbb{R}$$

59. $|z_1| = |z_2| = 1, z_1 + z_2 = 1$

$$a + bi + c + di = 1 \implies \begin{cases} a + c = 1 \\ b + d = 0 \end{cases} \implies \begin{cases} c = 1 - a \\ d = -b \end{cases}$$

$$\begin{cases} a^2 + b^2 = 1 \\ c^2 + d^2 = 1 \end{cases} \implies (1-a)^2 + (-b)^2 = 1 \implies a^2 + b^2 = 2a \implies a = \frac{1}{2} \implies c = \frac{1}{2}$$

$$\implies b^2 = 1 - \frac{1}{4} \implies b = \frac{\sqrt{3}}{2} \implies d = -\frac{\sqrt{3}}{2}$$

$$z_1 = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$z_2 = \frac{1}{2} - \frac{\sqrt{3}}{2}i$$

60. $z = \dfrac{(1+i)^2}{1-i} = \dfrac{2i}{1-i} \cdot \dfrac{1+i}{1+i} = \dfrac{2i-2}{2} = i-1$

$$\begin{cases} \rho = \sqrt{1^2+1^2} = \sqrt{2} \\ \cos\theta = \dfrac{\sqrt{2}}{2} \\ \sin\theta = -\dfrac{\sqrt{2}}{2} \end{cases} \implies \sqrt{2}(\cos\frac{7\pi}{4} + i\sin\frac{7\pi}{4})$$

61. **a)** $3+5i$

$$\begin{cases} \rho = \sqrt{3^2+5^2} = \sqrt{34} \\ \cos\theta = \dfrac{3\sqrt{34}}{34} \\ \sin\theta = \dfrac{5\sqrt{34}}{34} \end{cases} \implies \theta = \operatorname{atan}\frac{5}{3}$$

b) $-3+2i$

$$\begin{cases} \rho = \sqrt{3^2+2^2} = \sqrt{13} \\ \cos\theta = \dfrac{-3\sqrt{13}}{13} \\ \sin\theta = \dfrac{2\sqrt{13}}{13} \end{cases} \implies \theta = \operatorname{atan}\frac{-2}{3}$$

c) $-2-3i$

$$\begin{cases} \rho = \sqrt{2^2+3^2} = \sqrt{13} \\ \cos\theta = \dfrac{-2\sqrt{13}}{13} \\ \sin\theta = \dfrac{-3\sqrt{13}}{13} \end{cases} \implies \theta = \operatorname{atan}\frac{3}{2}$$

d) $1-4i$

$$\begin{cases} \rho = \sqrt{1^2+4^2} = \sqrt{17} \\ \cos\theta = \dfrac{\sqrt{17}}{17} \\ \sin\theta = \dfrac{-4\sqrt{17}}{17} \end{cases} \implies \theta = \operatorname{atan} -4$$

62. Dado $z_1 = \rho_1(\cos\theta_1 + i\sin\theta_1)$, $z_2 = \rho_2(\cos\theta_2 + i\sin\theta_2)$, prove que $|z_1+z_2| \le |z_1| + |z_2|$.

$z_1 + z_2 = (\rho_1\cos\theta_1 + \rho_2\cos\theta_2) + i(\rho_1\sin\theta_1 + \rho_2\sin\theta_2)$

$$|z_1+z_2| = \sqrt{(\rho_1^2\cos^2\theta_1 + 2\rho_1\rho_2\cos\theta_1\cos\theta_2 + \rho_2^2\cos^2\theta_2) + \ldots}$$
$$\overline{\ldots + (\rho_1^2\sin^2\theta_1 + 2\rho_1\rho_2\sin\theta_1\sin\theta_2 + \rho_2^2\sin^2\theta_2)}$$

$$= \sqrt{\rho_1^2(\cos^2\theta_1 + \sin^2\theta_1) + \rho_2(\cos^2\theta_2 + \sin^2\theta_2) + 2\rho_1\rho_2(\cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2)}$$

$$= \sqrt{\rho_1^2 + \rho_2^2 + 2\rho_1\rho_2\cos(\theta_1 - \theta_2)}$$

Como $\cos(\theta_1 - \theta_2)$ é no máximo 1, então:

$|z_1 + z_2| \leq \sqrt{(\rho_1 + \rho_2)^2} = |z_1| + |z_2|$

63. A soma de dois números representa a soma de dois vetores na origem, ou seja, um paralelogramo em que os vértices são: a origem, cada número na operação e o resultado da soma

64. **a)** $A = \{z \in \mathbb{C} : |z| = 2\}$- circunferência de raio 2 centrada na origem

b) $B = \{z \in \mathbb{C} : |z| \leq 3\}$- círculo de raio 3 centrada na origem

c) $C = \{z \in \mathbb{C} : |z - i| = 1\}$- circunferência de raio 1 centrada em $(0, 1)$

- O $-i$ move todo o plano para baixo, e para compensar os pontos devem mudar para cima

65. Dados $b = 2(\cos 30^\circ + i \sin 30^\circ)$ e $z = \rho(\cos\theta + i \sin\theta)$, qual a figura formada pelos afixos de $b, b + z, b + z + iz, b + iz$?

Podemos verificar geometricamente que para qualquer z e um ponto fixo b podemos criar um quadrado em que todos os lados tem lado de tamanho $|z|$.

Os números complexos que representam cada um dos lados desse quadrado são:

1. $b + z - b = z$
2. $b + z + iz - (b + z) = iz$
3. $b + z + iz - (b + iz) = z$
4. $b - (b + iz) = -iz$

Todos tem módulo z. Além disso podemos verificar que iz e z são perpendiculares e de mesmo módulo, o mesmo vale para os outros dois lados.

66. $z = x + yi, \dfrac{z}{\overline{z}} = -i$

$$z = -i\overline{z} \implies (x + yi) = -i(x - yi) \implies \begin{cases} x = -y \\ y = -x \end{cases}$$

$$\implies z = x - xi = x(1 - i) \iff y = -x$$

67. $\dfrac{z_1}{z_2} \in \mathbb{R}, z_1, z_2 \neq 0$

$$\frac{x + yi}{c + di} = r \implies \begin{cases} x = rc \\ y = rd \end{cases} \implies y = \frac{d}{c}x, \text{ uma reta qualquer que passe pela origem}$$

68. **a)** $A = \{z \in \mathbb{C} : \mathrm{Re}(z) = 0\}$- reta vertical que passa pela origem

b) $B = \{z \in \mathbb{C} : \mathrm{Im}(z) = 0\}$- reta horizontal que passar pela origem

c) $D = \{z \in \mathbb{C} : \text{Re}(z) \geq 1 \text{ e } \text{Im}(z) \geq 2\}$- área superior a direita limitada pelas retas $x = 2$ e $y = 1$

d) $E = \{z \in \mathbb{C} : |z| = 1\}$- circunferência de raio 1 centrado na origem

e) $F = \{z \in \mathbb{C} : |z| \leq 2\}$- círculo de raio 2 centrado na origem

69. $|z - (1 + i)| \geq 1$

O plano é deslocado 1 para baixo e 1 para a esquerda, então os pontos compensam, indo para $(1, 1)$.

É uma círculo de raio 1 centrado em $(1, 1)$.

70. $t = 2 + 3i, A = \{z \in \mathbb{C} : |z - t| \leq 1\}, B = \{z \in \mathbb{C} : z = a + bi, b \leq 3\}, A \cap B = ?$

Metade de baixo do círculo de raio 1 centrado em $(2, 3)$.

71. Fixado $\theta, \rho(\cos\theta + i \sin\theta)$

Pertence a reta com esse ângulo com eixo, ρ é como um fator de escala que mapeia a reta.

72. **a)** $\text{Re}(w) = u, \text{Im}(w) = v, w = 1 - \dfrac{1}{z}, z = x + yi$

$$w = \frac{z - 1}{z} \cdot \frac{\overline{z}}{\overline{z}} = \frac{|z| - \overline{z}}{|z|^2} = \frac{x^2 + y^2 - x + yi}{x^2 + y^2}$$

$$(u, v) = \left(\frac{x^2 + y^2 - x}{x^2 + y^2}, \frac{y}{x^2 + y^2}\right)$$

b) $x = y$

$$\begin{cases} u = \dfrac{2x^2 - x}{2x^2} = \dfrac{2x - 1}{2x} \\ v = \dfrac{x}{2x^2} = \dfrac{1}{2x} \end{cases} \implies u = v(\frac{1}{v} - 1) = 1 - v$$

73. $z : \theta$ é o menor argumento, para $|z - 25i| \leq 15$

$|z - 25i| \leq$ representa um círculo de raio 15 centrado em $(0, 25)$. Representando graficamente, o número complexo com menor argumento será a reta que passa na origem e tangencia a circunferência, formando o triângulo retângulo com pontos $(0, 0), z_0 = (x, y), (0, 25)$ e com lados z_0, 15 e 25.

Podemos usar pitágoras para calcular o módulo de z_0 : $25^2 = 15^2 + |z_0|^2 \implies |z_0| = 20$

E podemos calcular o seno e cosseno de θ usando a própria definição:

$$\begin{cases} \cos\left(\dfrac{\pi}{2} - \theta\right) = \sin\theta = \dfrac{20}{25} = \dfrac{4}{5} \\ \sin\left(\dfrac{\pi}{2} - \theta\right) = \cos\theta = \dfrac{15}{25} = \dfrac{3}{5} \end{cases} \implies z_0 = 20\left(\frac{3}{5} + \frac{4}{5}i\right) = 12 + 16i$$

74. Dado $z = \rho(\cos\theta + i\sin\theta)$, prove que $\operatorname{Im}\left(\frac{z}{z^2+\rho^2}\right) = 0, \operatorname{Re}\left(\frac{\rho - iz}{\rho + iz}\right) = 0$

$$\frac{z}{z^2+\rho^2}\cdot\frac{\overline{z^2}+\rho^2}{\overline{z^2}+\rho^2} = \frac{(z\overline{z})\overline{z}+z\rho^2}{z^2\overline{z^2}+\rho^2z^2+\rho^2\overline{z^2}+\rho^4} = \frac{\rho^2\overline{z}+\rho^2z}{\rho^4+\rho^2(z^2+\overline{z}^2)+\rho^4} = \frac{2\rho^2\operatorname{Re}(z^2)}{2\rho^2(\rho^2+\operatorname{Re}(z^2))}$$

$$= \frac{\operatorname{Re}(z)}{\rho^2+\operatorname{Re}(z^2)} \implies \in \mathbb{R}$$

$$= \frac{\rho-iz}{\rho+iz}\cdot\frac{\rho-i\overline{z}}{\rho-i\overline{z}} = \frac{\rho^2-i\rho\overline{z}-i\rho z+i^2z\overline{z}}{\rho^2-i\overline{z}\rho+iz\rho-i^2z\overline{z}} = \frac{-i\rho(z+\overline{z})}{2\rho^2-i\rho(\overline{z}-z)} = \frac{-2i\operatorname{Re}(z)}{2\rho-i(-2i\operatorname{Im}(z))}$$

$$= \frac{-i\operatorname{Re}(z)}{\rho+\operatorname{Im}(z)} \implies \in \mathbb{C}-\mathbb{R}$$

75. $\dfrac{(a+bi)^2}{c+di} - \dfrac{1}{a+ci} - \dfrac{1+i}{13i}$

$|a+bi| = 5, \theta_{ab} \in [0, \frac{\pi}{2}], b-a = 1, (c+di)^2 = -5-12i, c < 0$

$$b-a=1 \implies b^2-2ab+a^2 = 1 \implies (a^2+b^2)-2ab = 1 \implies 5^2-2ab = 1 \implies ab = 12$$

$$a^2+\left(\frac{12}{a}\right)^2 = 5^2 \implies (a^2)^2-5^2(a^2)+12^2 = 0 \implies a^2 = \frac{5^2 \pm \sqrt{5^4-(2\cdot 12)^2}}{2}$$

$$\implies a^2 = \frac{25\pm 7}{2} \implies a^2 = \begin{cases}16\\9\end{cases} \implies \begin{cases}a=4 \implies b=3\\a=3 \implies b=4\end{cases} \underset{b-a=1}{\implies} \begin{cases}a=3\\b=4\end{cases}$$

$$(c^2-d^2, 2cd) = (-5,-12) \implies \begin{cases}d^2-c^2=5\\cd=-6\end{cases} \implies d^2-(\tfrac{-6}{d})^2 = 5 \implies d^4-6^2 = 5d^2$$

$$(d^2)^2-5(d^2)-6^2 = 0 \implies d^2 = \frac{5\pm\sqrt{5^2+(2\cdot 6)^2}}{2} = \frac{5\pm 13}{2} \implies \begin{cases}d^2=-4 \implies \perp\\d=\pm 3\end{cases}$$

$$c = \frac{-6}{d} \implies \begin{cases}\frac{-6}{3}-2\\\frac{-6}{-3}=2\end{cases} \underset{c<0}{\implies} \begin{cases}c=-2\\d=3\end{cases}$$

$$\frac{(3+4i)^2}{-2+3i} - \frac{1}{3-2i} - \frac{1+i}{13i} = \frac{3^2-4^2+2\cdot 3\cdot 4i}{-2+3i} - \left(\frac{13i+(3,-2)(1,1)}{13(3i+2)}\right)$$

$$= \frac{9-16+24i}{(-2,3)} - \frac{(3+2,3-2)+(0,13)}{13(2,3)} = \frac{(-7,24)}{(-2,3)} - \frac{(5,14)}{13(2,3)}$$

$$= \frac{13(2,3)(-7,24)-(-2,3)(5,14)}{13(2,3)(-2,3)} = \frac{13(-14-72,48-21)-(-10-42,-28+15)}{13(-4-9,6-6)}$$

$$= \frac{13(-86,27)-(-52,-13)}{13(-13,0)} = \frac{13(-86,27)+13(4,1)}{-13^2} = \frac{(-86+4,27+1)}{-13} = \frac{82-28i}{13}$$

76. $z = 1+i$

$$|z^4| = |z|^4 = |1+i|^4 = (\sqrt{2})^4 = 2^2 = 4$$

$$\begin{cases}\cos\theta = \frac{\sqrt{2}}{2} \\ \sin\theta = \frac{\sqrt{2}}{2}\end{cases} \implies \theta = \frac{\pi}{4} \underset{\theta_{z^4}=4\theta}{\implies} \theta_{z^4} = \pi$$

77. **a)** $\left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)^{100}$

$$\begin{cases}\rho = \sqrt{\frac{1}{4} + \frac{3}{4}} = 1 \\ \cos\theta = -\frac{1}{2} \\ \sin\theta = \frac{\sqrt{3}}{2}\end{cases} \implies \theta = \frac{2\pi}{3} \implies (\cos\frac{200\pi}{3} + i\sin\frac{200\pi}{3})$$

b) $(3-3i)^{-12}$

$$\begin{cases}\rho = \sqrt{3^2+3^2} = 3\sqrt{2} \\ \cos\theta = \frac{\sqrt{2}}{2} \\ \sin\theta = -\frac{\sqrt{2}}{2}\end{cases} \implies \theta = \frac{7\pi}{4} \implies (3\sqrt{2})^{-12}(\cos\left(\frac{-12\cdot 7\pi}{4}\right) + i\sin\left(\frac{-12\cdot 7\pi}{4}\right))$$

$$= \frac{1}{3^{12}\sqrt{2^{12}}}(\cos(-21\pi) + i\sin(-21\pi)) = \frac{-1}{3^{12}\cdot 2^6}$$

c) $(-\sqrt{3}-i)^{20}$

$$\begin{cases}\rho = \sqrt{\sqrt{3}^2+1^2} = \sqrt{4} = 2 \\ \cos\theta = -\frac{\sqrt{3}}{2} \\ \sin\theta = -\frac{1}{2}\end{cases} \implies \theta = \frac{7\pi}{6} \implies 2^{20}(\cos\left(\frac{20\cdot 7\pi}{6}\right) + i\sin\left(\frac{20\cdot 7\pi}{6}\right))$$

$$= 2^{20}(\cos(\frac{70\pi}{3}) + i\sin(\frac{70\pi}{3}))$$

78. ...

79. $\sin 2\theta, \cos 2\theta$

$$(\cos\theta + i\sin\theta)^2 = \cos 2\theta + i\sin 2\theta$$

$$\cos^2\theta - \sin^2\theta + 2i\cos\theta\sin\theta = \cos 2\theta + i\sin 2\theta$$

$$\begin{cases}\cos 2\theta = \cos^2\theta - \sin^2\theta \\ \sin 2\theta = 2\sin\theta\cos\theta\end{cases}$$

80. $\sin 3\theta, \cos 3\theta$

$$(\cos\theta + i\sin\theta)^3 = \cos 3\theta + i\sin 3\theta$$

$$\cos^3\theta + 3i\cos^2\theta\sin\theta - 3\cos\theta\sin^2\theta - i\sin^3\theta = \cos 3\theta + i\sin 3\theta$$

$$\begin{cases}\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta \\ \sin 3\theta = 3\cos^2\theta\sin\theta - \sin^3\theta\end{cases}$$

81. $\min n \in \mathbb{N} | (i-\sqrt{3})^n \in \mathbb{C} - \mathbb{R}$

$$\begin{cases}\rho = \sqrt{1^2 + \sqrt{3}^2} = 2 \\ \cos\theta = -\dfrac{\sqrt{3}}{2} \\ \sin\theta = \dfrac{1}{2}\end{cases} \implies \theta = \frac{5\pi}{6} \implies 2^n\left(\cos\left(\frac{5n\pi}{6}\right) + i\sin\left(\frac{5n\pi}{6}\right)\right)$$

$$\implies \cos\frac{5n\pi}{6} = 0 \implies \frac{5n\pi}{6} = \frac{\pi}{2} + k\pi \implies 5\cdot\frac{n}{3} = 1 + 2k \implies n = 3$$

82. **a)** $\min n \in \mathbb{N}, (\sqrt{3}+i)^n \in \mathbb{R}, z > 0$

$$\begin{cases}\rho = \sqrt{\sqrt{3}^2 + 1} = 2 \\ \cos\theta = \dfrac{\sqrt{3}}{2} \\ \sin\theta = \dfrac{1}{2}^2\end{cases} \implies \theta = \frac{\pi}{6}$$

$$2^n\left(\cos\frac{n\pi}{6} + i\sin\frac{n\pi}{6}\right)$$

$$\cos\frac{n\pi}{6} = 1 = \cos 0 \implies \frac{n\pi}{6} = 2k\pi \implies n = 12k \implies n = 0$$

b) $\min n \in \mathbb{N}, (\sqrt{3}+i)^n \in \mathbb{R}, z < 0$

$$\cos\frac{n\pi}{6} = -1 = \cos\pi \implies \frac{n\pi}{6} = \pi + 2k\pi \implies n = 6(2k+1) \implies n = 6$$

c) $\min n \in \mathbb{N}, (\sqrt{3}+i)^n \in \mathbb{C} - \mathbb{R}$

$$\cos\frac{n\pi}{6} = 0 = \cos\frac{\pi}{2} \implies \frac{n\pi}{6} = \frac{\pi}{2} + k\pi \implies n = 3(2k+1) \implies n = 3$$

83. $iz + 2\overline{z} + 1 - i = 0$

a) $i(a+bi) + 2(a-bi) + (1-i) = 0 \implies ai - b + 2a - 2bi + 1 - i = 0$

$$\implies \begin{cases}-b + 2a + 1 = 0 \\ a - 2b - 1 = 0\end{cases} \implies \begin{cases}2a - b = -1 \\ a - 2b = 1\end{cases}$$

$$\implies \begin{cases}a = -1 \\ b = -1\end{cases} \implies z = -(1+i)$$

b) $$\begin{cases}\rho = \sqrt{2} \\ \theta = \dfrac{5\pi}{4}\end{cases}$$

c) $z^{1004} = (\sqrt{2})^{1004}\left(\cos\left(\frac{1004\cdot 5\pi}{4}\right) + i\sin\left(\frac{1004\cdot 5\pi}{4}\right)\right)$

$= 2^{502}(\cos(251 \cdot 5\pi) + i \sin(251 \cdot 5\pi)) = -2^{502}$

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100. ...

101. ...

102. **a)** $x^2 - i = 0$

$$x = \sqrt{i} \implies x = (\cos\left(\frac{2k\pi + \frac{\pi}{2}}{2}\right) + i \sin\left(\frac{2k\pi + \frac{\pi}{2}}{2}\right)) = \begin{cases} \frac{\sqrt{2}}{2}(1+i) \\ -\frac{\sqrt{2}}{2}(1+i) \end{cases}$$

b) $x^6 + 8 = 0$

$$x = \sqrt[6]{-8} \implies x = \sqrt{2}(\cos\frac{\pi}{6}(2k+1) + i\sin\frac{\pi}{6}(2k+1)) = \begin{cases} \frac{\sqrt{6}}{2} + i\frac{\sqrt{2}}{2} \\ i\sqrt{2} \\ -\frac{\sqrt{6}}{2} + i\frac{\sqrt{2}}{2} \\ -\frac{\sqrt{6}}{2} - i\frac{\sqrt{2}}{2} \\ -i\sqrt{2} \\ \frac{\sqrt{6}}{2} - i\frac{\sqrt{2}}{2} \end{cases}$$

c) $x^4 - 1 + i = 0$

$$x = \sqrt[4]{1-i} \implies x = \sqrt{2}(\cos\left(\frac{\pi}{16}(8k+7)\right) + i\sin\left(\frac{\pi}{16}(8k+7)\right))$$

d) $x^3 + 1 = 0$

$$x = \sqrt[3]{-1} \implies x = 1(\cos\frac{\pi}{3}(2k+1) + i\sin\frac{\pi}{3}(2k+1)) = \begin{cases} \frac{1}{2} + i\frac{\sqrt{3}}{2} \\ -1 \\ \frac{1}{2} - i\frac{\sqrt{3}}{2} \end{cases}$$

e) $x^4 + i = 0$

$$x = \sqrt[4]{-i} \implies x = 1(\cos\frac{\pi}{8}(4k+3) + i\sin\frac{\pi}{8}(4k+3))$$

f) $x^3 - 27 = 0$

$$x = \sqrt[3]{27} \implies x = 27(\cos\frac{2k\pi}{3} + i\sin\frac{2k\pi}{3}) = \begin{cases} 27 \\ \dots \\ \dots \end{cases}$$

103. a) $x^8 - 17x^4 + 16 = 0$

$$y = x^4 \implies y^2 - 17y + 16 = 0 \implies y = \frac{17 \pm \sqrt{17^2 - 4\cdot 16}}{2} = \frac{17 \pm 15}{2} = \begin{cases} 16 \\ 1 \end{cases}$$

$$\implies \begin{cases} x^4 = 16 \implies x = 2(\cos\frac{k\pi}{2} + i\sin\frac{k\pi}{2}) \in \{2, 2i, -2, -2i\} \\ x^4 = 1 \implies x = 1(\cos\frac{k\pi}{2} + i\sin\frac{k\pi}{2}) \in \{1, i, -1, -i\} \end{cases}$$

b) $x^6 + 9x^3 + 8 = 0$

$$y = x^3 \implies y^2 + 9y + 8 = 0 \implies y = \frac{-9 \pm \sqrt{81 - 4\cdot 8}}{2} = \frac{-9 \pm 7}{2} = \begin{cases} -8 \\ 1 \end{cases}$$

$$\implies \begin{cases} x^3 = -8 \implies x = 2(\cos\frac{2k\pi+\pi}{3} + i\sin\frac{2k\pi+\pi}{3}) \in \{1 + i\sqrt{3}, -2, 1 - i\sqrt{3}\} \\ x^3 = -1 \implies x = 1(\cos\frac{2k\pi+\pi}{3} + i\sin\frac{2k\pi+\pi}{3}) \in \{\frac{1}{2} + i\frac{\sqrt{3}}{2}, -1, \frac{1}{2} - i\frac{\sqrt{3}}{2}\} \end{cases}$$

c) $x^4 - 2x^2 + 2 = 0$

$$y = x^2 \implies y^2 - 2y + 2 = 0 \implies y = \frac{2 \pm \sqrt{4 - 4\cdot 2}}{2} = \frac{2 \pm 2i}{2} = \begin{cases} 1 + i \\ 1 - i \end{cases}$$

$$\implies \begin{cases} x^2 = 1+i \implies x \in \{(\cos\frac{\pi}{8} + i\sin\frac{\pi}{8}), (\cos\frac{9\pi}{8} + i\sin\frac{9\pi}{8})\} \\ x^2 = 1-i \implies x \in \{(\cos\frac{7\pi}{8} + i\sin\frac{7\pi}{8}), (\cos\frac{15\pi}{8} + i\sin\frac{15\pi}{8})\} \end{cases}$$

d) $x^4 - 5x^2 + 4 = 0$

$$y = x^2 \implies y^2 - 5y + 4 = 0 \implies y = \frac{5 \pm \sqrt{25 - 4\cdot 4}}{2} = \frac{5 \pm 3}{2} = \begin{cases} 4 \\ 1 \end{cases}$$

$$\implies \begin{cases} x^2 = 4 \implies x \in \{2, -2\} \\ x^2 = 1 \implies x \in \{1, -1\} \end{cases}$$

e) $27x^6 + 35x^3 + 8 = 0$

$$y = x^3 \implies 27y^2 + 35y + 8 = 0 \implies y = \frac{-35 \pm \sqrt{35^2 - 4\cdot 27\cdot 8}}{2\cdot 27} = \frac{-35 \pm 19}{54} = \begin{cases} -1 \\ \frac{8}{27} \end{cases}$$

$$\implies \begin{cases} x^3 = -1 \implies x \in \{-1, \frac{1}{2} + i\frac{\sqrt{3}}{2}, \frac{1}{2} + i\frac{\sqrt{3}}{2}\} \\ x^3 = \frac{8}{27} \implies x \in \{\frac{2}{3}, -\frac{1}{3} + i\frac{\sqrt{3}}{3}, -\frac{1}{3} - i\frac{\sqrt{3}}{3}\} \end{cases}$$

f) $ix^2 - 2x + \sqrt{3} = 0$

$$x = \frac{2 \pm \sqrt{4 - 4i\sqrt{3}}}{2i} = \frac{2 \pm 2\sqrt{1 - i\sqrt{3}}}{2i} \cdot \frac{-i}{-i} = -i(1 \pm \sqrt{1 - i\sqrt{3}})$$

$$\sqrt{1 - i\sqrt{3}} = \pm\left(\frac{\sqrt{6}}{2} - i\frac{\sqrt{6}}{2}\right)$$

$$= -i\left(1 + \frac{\sqrt{6}}{2} - i\frac{\sqrt{2}}{2}\right) = \frac{\sqrt{2}}{2} - i\left(\frac{\sqrt{6}}{2} + 1\right)$$

$$= -i\left(1 + \frac{\sqrt{6}}{2} + i\frac{\sqrt{2}}{2}\right) = -\frac{\sqrt{2}}{2} + i\left(\frac{\sqrt{6}}{2} - 1\right)$$